

NETWORKING 06

HTTP vs HTTPS

👉 HTTP = communication between browser and website

📌 Simple:

HTTP = sending data in **plain text**

❌ Problem:

- Anyone can read data 😬
- Not secure

💡 Example:

You login:

```
username: admin  
password: 1234
```

➡ Hacker can see this easily

WHAT IS HTTPS?

👉 HTTPS = secure version of HTTP

📌 Simple:

HTTPS = **encrypted communication**

🔒 Uses:

- SSL / TLS encryption

💡 Example:

```
username: admin  
password: 1234
```

➡ Becomes:

```
XJ7#@!K$%...
```

➡ Hacker sees only garbage

Feature	HTTP	HTTPS
Security	❌ Not secure	✅ Secure
Data	Plain text	Encrypted
Use	Old / unsafe	Modern / safe
Port	80	443

FIRST: WHAT IS A SERVER?

Simple meaning:

 **Server = powerful computer that stores websites/data**

 Example:

- You open [google.com](https://www.google.com)
- Your browser sends request
- Google server sends response

Where is server located?

 Usually in:

- Data centers 
- Big companies (Google, AWS, etc.)
- Cloud (not your local PC)

1 2
3 4

NOW STATUS CODES (DEEP UNDERSTANDING)



403 FORBIDDEN (IMPORTANT)



Meaning:

Server understood your request BUT refuses to allow you



WHY DOES 403 HAPPEN?

1

No permission

👉 Example:

- Admin page

```
/admin
```

→ Only admin allowed

→ You are normal user → ❌ 403

2

IP blocked 🚫

👉 Server blocks:

- Your country
- Your IP
- Suspicious traffic

→ You get 403

3

Missing authentication

👉 You didn't login

→ Server says: "No access"

4

File permissions (server side)

👉 Server file is restricted

→ Even if exists, you cannot access



Forensic thinking:

👉 Many 403 requests =

→ Possible attacker scanning pages

404 NOT FOUND

Meaning:

Resource does not exist

Reasons:

- Wrong URL
- Page deleted
- Typo

Cyber thinking:

👉 Too many 404 =

➡ Directory brute force attack

500 SERVER ERROR (VERY IMPORTANT)

Meaning:

Problem inside server (not your fault)

What type of problems?

1 Code error

👉 Website code crashed

➡ Bug in backend

2 Database failure

👉 Server cannot fetch data

3 Server overload

👉 Too many users

➡ Server crashes

4 Misconfiguration

👉 Server settings wrong

Forensic thinking:

👉 Repeated 500 errors =

➡ Possible:

- Attack
- Misconfiguration
- Exploit attempt



301 REDIRECT



Meaning:

Page moved permanently

Example:

`http://site.com → https://site.com`



Automatically redirected



Cyber use:



Attackers may use fake redirects for phishing



200 OK



Meaning:

Everything worked fine



But still:



Even 200 can be dangerous



If fake website returns 200 😈